

Problem H. K-th Number

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

[English](#)
[Vietnamese](#)

You are working for Macrohard company in data structures department. After failing your previous task about key insertion you were asked to write a new data structure that would be able to return quickly k -th order statistics in the array segment.

That is, given an array $a[1 \dots n]$ of different integer numbers, your program must answer a series of questions $Q(i, j, k)$ in the form: "What would be the k -th number in $a[i \dots j]$ segment, if this segment was sorted?"

For example, consider the array $a = (1, 5, 2, 6, 3, 7, 4)$. Let the question be $Q(2, 5, 3)$. The segment $a[2 \dots 5]$ is $(5, 2, 6, 3)$. If we sort this segment, we get $(2, 3, 5, 6)$, the third number is 5, and therefore the answer to the question is 5.

Input

The first line of the input contains n — the size of the array, and m — the number of questions to answer ($1 \leq n \leq 100000$, $1 \leq m \leq 5000$).

The second line contains n different integer numbers not exceeding 10^9 by their absolute values — the array for which the answers should be given.

The following m lines contain question descriptions, each description consists of three numbers: i, j , and k ($1 \leq i \leq j \leq n$, $1 \leq k \leq j - i + 1$) and represents the question $Q(i, j, k)$.

Output

For each question output the answer to it — the k -th number in sorted $a[i \dots j]$ segment.

Example

Input	Output
7 3 1 5 2 6 3 7 4 2 5 3 4 4 1 1 7 3	5 6 3

Note: a naive solution will not work!!!